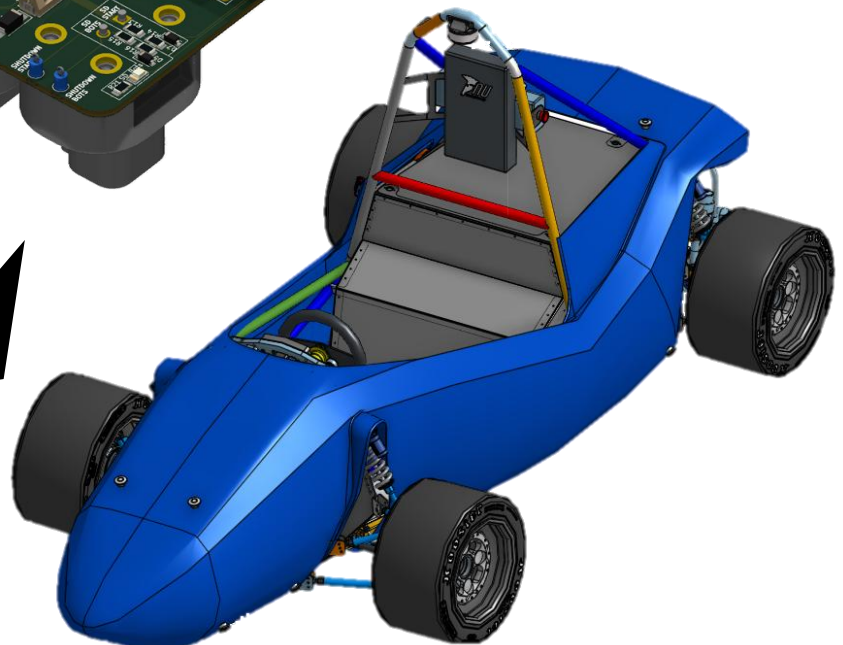
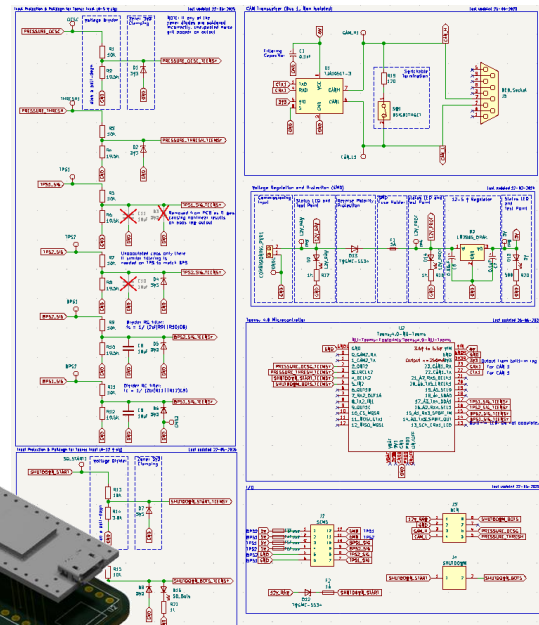
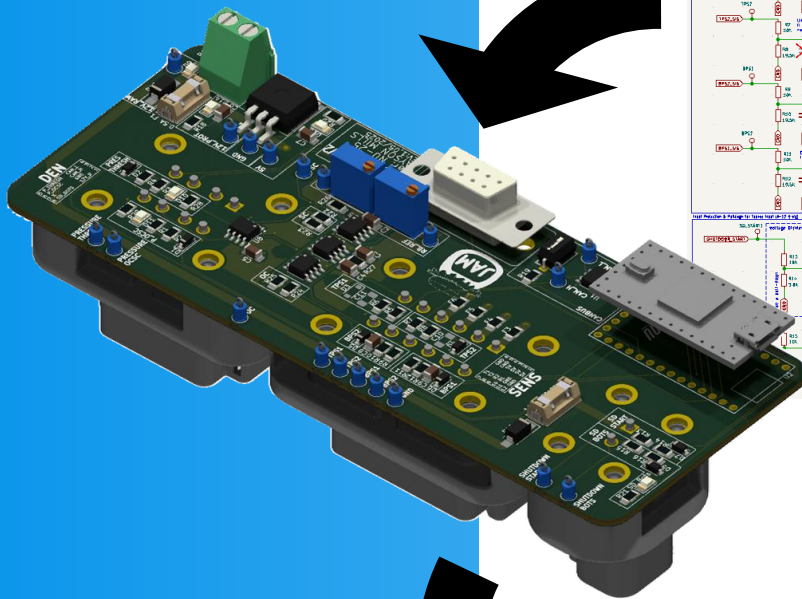




How to Create a PCB for NU Racing

ZERO-TO-HERO



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2025



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Introduction

This Zero-to-Hero is intended for NU Racing engineers creating their first PCB for the competition platform. It builds upon the skills developed in the PCB Design Zero-to-Hero introducing more team-specific techniques unique to working within the team. It is mandatory to complete PCB Zero-to-Hero and its advanced modules prior to contributing towards the car. When designing and building parts for the competition platform, higher standards must be upheld in comparison to junior projects. As a result, this well-defined design process and workflow has been developed over the history of the team for you to take advantage of.

How to Use This Document:

- Follow the modules sequentially and refer to each section as you progress through the PCB life cycle. Understand each section prior to completing the corresponding stage in design. It is unnecessary to absorb the whole document at once. Instead engage with sections when applicable.
- Create a copy of the PCB Design Checklist and complete it as you progress.
- Justification is required when deviating from the processes and methods accepted by NU Racing and described in this document.
- To address gaps in understanding, refer to the relevant Zero-to-Hero modules and seek guidance from experienced NU Teams engineers.



PCB Design Checklist

Preliminary Research

- ☐ Read previous reports & DDRs
- ☐ Review FSAE rules and addendum
- ☐ Learn the topology of the system
- ☐ Refer to Electrical Systems Overview
- ☐ Review Lessons Learnt document

Schematic

- ☐ Use standard circuits
- ☐ Validate nonstandard circuits
- ☐ Supply data sheets
- ☐ Check rules compliance
- ☐ Polish schematic
- ☐ Complete DDR PCB Design Intent, checked by MCHA lead
- ☐ Complete DDR Schematic Review, checked by MCHA lead

Layout

- ☐ Check footprint assignment with 1:1 printed default layout
- ☐ Place connectors and set edge cuts
- ☐ Place functional groups and draft layout
- ☐ Set critical traces and define fill zones
- ☐ Lay traces and refine part placement
- ☐ Polish the Silkscreen, label functional groups and connector pinouts
- ☐ Complete DDR Layout Review, checked by MCHA lead
- ☐ Export STEP file of PCB and 3D print for MECH department
- ☐ Complete DDR CAD and Physical Design Review, checked by MCHA lead

Prepare for Order

- ☐ Full DDR checked by MCHA Department Lead, Chief Engineer, and Malcolm Sidney
- ☐ Complete BOM
- ☐ Order PCB parts
- ☐ Order PCB

Commissioning

- ☐ Prepare commissioning procedure and unit test checklist
- ☐ Manufacture and unit test PCB
- ☐ Integrate with MCHA systems
- ☐ Complete track day

Maintenance

- ☐ Maintain documentation
- ☐ Track issues and make recommendations
- ☐ Possibly iterate the design

NU Racing Design Rationale

The rationale behind this design process is to **minimise risk** and maximise the chance of success for both you and the team. This is done by **validating every step** of the design process and keeping yourself **accountable** by checking every aspect of your design and leaving nothing to chance or intuition.

The most critical thing is to have a **finished, working and rules compliant** car by the end of the design cycle and before competition. As a result, the following criteria need to be considered at every stage:

- **Reliability** – is the solution likely to fail or will it work consistently in expected use cases? Aim to be completely confident in the designs ability to always perform its minimum functionality. *“To finish first, first you have to finish”.*
- **Cost** – Avoid unnecessary spending and be aware of the cost effectiveness of other solutions. NU Teams has a budget to maintain, and significant spending should only occur when critical (for PCBs, this mainly applies to component selection).
- **Ease of compliance** – How easily does the solution conform to the FSAE rules and can be proven in tech inspection? Do not operate in grey areas and ambiguous rule interpretation as this risks failing tech inspection.
- **Time** – The PCB must be designed, manufactured and commissioned within project timelines. Be aware of others depending on the completion of your PCB and by being risk adverse, minimise the chance of delaying team progress. *Spending time being methodical early can save large amounts troubleshooting later!*
- **Serviceability** – Given an issue, troubleshooting, fixing and servicing should be straightforward. Aim to make issue sensitive parts of the design more accessible and faults easy to diagnose.
- **Simplicity** – Often a result of the above criteria, is the design as simple as possible? The solution should not exceed the knowledge and skill level of the team.
- **Effectiveness** – Is the problem solved? In isolation, is the desired function achieved by the system?
- **Performance** – Aim to increase on track performance and improve energy efficiency of the car. By considering the other criteria effectively, the overall performance of the team operationally is also impacted.

To help judge against these criteria, refer to the **NU Racing decision matrix** by Alec Chapman (*NU Teams SharePoint -> Documents -> NU Racing*).

Documentation:

NU Racing highly values consistent **documentation** and emphasises **knowledge transfer**. It is imperative that you document (and make available to the team) as much as possible. This will not only strengthen your understanding of your project and serve as a memory aid for yourself but will also allow other team members to stay up to date with your progress and future members to get up to speed. *Realise how important other engineers' documentation has been for you and pay it forward!* This is to **ensure no information is lost** in your absence and that others can integrate with your system without relying on your direct involvement.

Keep these ideals in mind throughout the design cycle.

Preliminary Research

Before diving into design, **preliminary research is crucial** to ensure all **system requirements** are fully understood. Please note that these steps may vary depending on if you are making a new revision of a proven design or creating a brand-new PCB.

Rules and compliance:

- Learn to navigate the **FSAE rules** and following addendums (<https://www.saea.com.au/rules-documents-templates> or Discord -> NU Racing -> templates-and-references-documents).
- **Stay up to date** with new addendums as they are added multiple times per year.
- Find all the **rules that relate to your system** and PCBs in general and **understand them** (*you may find it beneficial to copy them into a separate document for easy access*)
- **Check in with these regularly** and communicate with your department lead about any aspects of tech inspection and compliance you don't understand.

Literature Review

- All pre-existing reports relevant to your PCB should be made available to you. (Discord -> NU Racing -> fyp-docs or via contact with team leads)
- **Review the report** sections pertaining to your PCB. It is not necessary to memorise everything but important to know where to find information. You may choose to summarise the report or take notes to consolidate understanding.
- Be aware of conclusions, **recommendations** and future work identified in previous reports as any within your scope can improve performance.
- Recommendations and previously identified issues may also be located within the **previous** Unit Test Checklist and Commissioning Notes tab within the PCBs DDR.
- While reading the reports and documentation, **familiarise yourself with the topology** of the system by cross examining with the KiCad schematic (*note how functional blocks improve usability of the schematic*). Previous DDRs also contains information on system requirements and design changes in the PCB Design Intent section.
- Refer to the **electrical systems overview** (NU-Teams GitHub -> Racing-NU2X -> NU2X Electrical Systems Overview) for in-depth integration information on the MCHA systems.
- Also available is the **NU Racing Lessons Learnt** document developed by Alec Chapman containing problem overviews and recommendations for many systems on the car (NU Teams SharePoint -> Documents -> NU Racing. Note this document is a work in progress)
- To strengthen conceptual knowledge, consider creating a flow chart of how signals are processed across the board or write summary of what each functional block does and how.

PCB Design

When making design decisions and changes, every choice must be justified, validated, and documented. Before working on the PCB files, familiarise with the **DDR requirements** to avoid unnecessary revisions. Also develop good habits when interacting with the NU Teams **GitHub** and racing repository you are working from. Always git **pull origin** prior to working on any files and git **push** with a commit after a session of work. Provide clear, concise comments summarising changes made when pushing. These comments do not have to be written just before pushing your work. Good practice is to write these comments as you make the changes (GitHub will remember what you have written). These comments allow the team to track when and what edits were made, avoid miscommunication, and allows version control to revert changes.

Track any changes made to the system by leveraging the MCHA design changes document (*NU Teams Sharepoint -> Documents -> NU Racing -> Electric Vehicle -> 202X -> MCHA -> 202X MCHA Systems Design Changes*)

Schematic:

- Where possible, use **standard circuits** (*NU-Teams GitHub -> NU-Teams-KiCAD-Library -> Standard-Circuits*). These standard blocks are reliable, proven designs that minimise risk by ensuring racing PCBs use homologated topology where possible and improves system integration. For example, incorrectly implementing CAN, using incompatible connectors or changing important pinout arrangements will render your PCB incompatible with the competition platform. For more detail on the standard circuits and components, refer to *"Daniel_Iveson_FYP_Report" section 5.2 (Discord -> fyp-docs)*.
- This also applies to validated circuit designs not found in the standard blocks but applied to other PCBs. For example, it may be useful to copy the HV detection block from the AIL or the latching relay design in the HFR rather than developing an unproven alternative.
- When creating bespoke **nonstandard circuits, calculations** must sufficiently prove the intended purpose of the design. It must then be **simulated** using a tool such as LTSpice or Logisim. This may be extended to the use of **bread board prototyping** especially when using non regular components. The results of these methods must be unanimous in validating the concept.
- Data sheets must be supplied for **nonstandard components**. At minimum, utilise the data sheet field in the symbol properties menu (*data sheets may also be included in the SharePoint or "Datasheets" folder within the PCB GitHub file path*). Be certain to check **voltage and current ratings** as well as the **rules compliance** of the part. Be wary of correct footprint and **pinout arrangement** of nonstandard parts when assigning symbols and footprints.
- Consider any Teensy pin requirements. Ensure correct use of analogue vs digital pins and additional functions such as CAN, PWM, I2C, SPI etc. Be aware of libraries your system's code uses which may interfere with certain pins. Issues arise when incorrectly implementing pins. For example, if an analogue signal is connected to a PWM (analogue write pin), the teensy will not be able to read it.
- When polishing your schematic, continuously refer to the DDR fields. Aim to **abstract away complexity** with the use of functional blocks, comments and legends (*see figure1 for examples*).

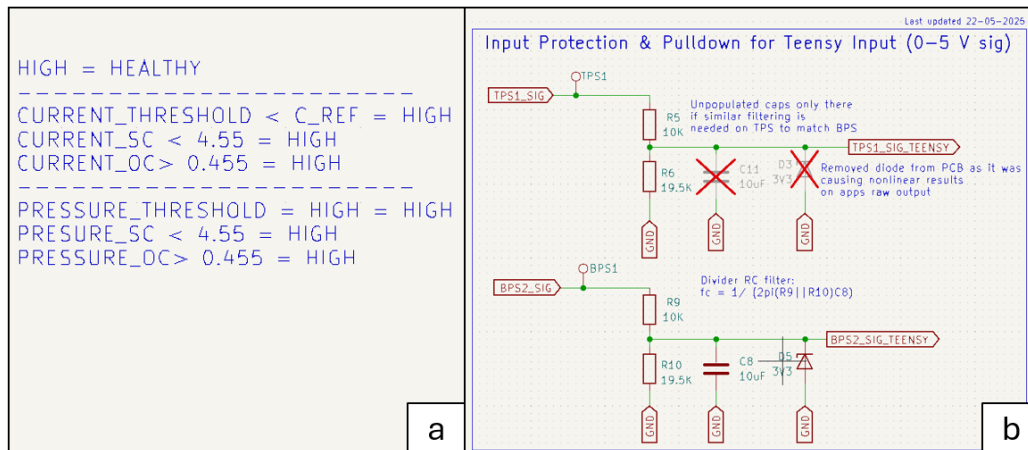


Figure 1a: BSPD logic legend. 1b: PEN commenting and block implementation

- To validate footprint assignment, print a 1:1 scale copy of the default PCB layout using *File -> Print -> *select important layers such as copper and silkscreen* -> Scale 1:1 -> *print to PDF or to a printer**. For dual sided PCBs, print front and back copper separately. Check footprints by placing the physical parts on the paper footprint pads. Also confirm the pinout of the exact part IDs stocked in the MCHA lab.
- Before commencing the layout, have the schematic DDR sheet checked. Each DDR section should be checked by the MCHA Department Lead as they are completed and prior to moving on to the next stage of design. This avoids backtracking.

Layout

- Once again, with DDR demands in mind, begin by grouping functional blocks and considering space/ **edge cut requirements**. Place critical receptacles first, ensuring they are accessible then place larger component where they best fit. Spend time logically laying out component groups and think ahead to trace layout for a cohesive design. Many layout iterations are often needed to balance trace route complexity with part placement.
- **Set critical traces** early, such as power lines, shutdown circuit and sensitive analogue signals. It may be beneficial to consider these before laying out all components. However, leave the rest of trace routing until after part placement to avoid rerouting. Effective trace routing is illustrated by Figure 2.
- **Trace requirements** to consider:
 - Shutdown circuit compliance and power lines ampacity (e.g. 2mm trace thickness)
 - Can_H and Can_L length matching and proximity routing
 - Avoid placing vias on analogue signals
 - Do **not** place vias on solder pads

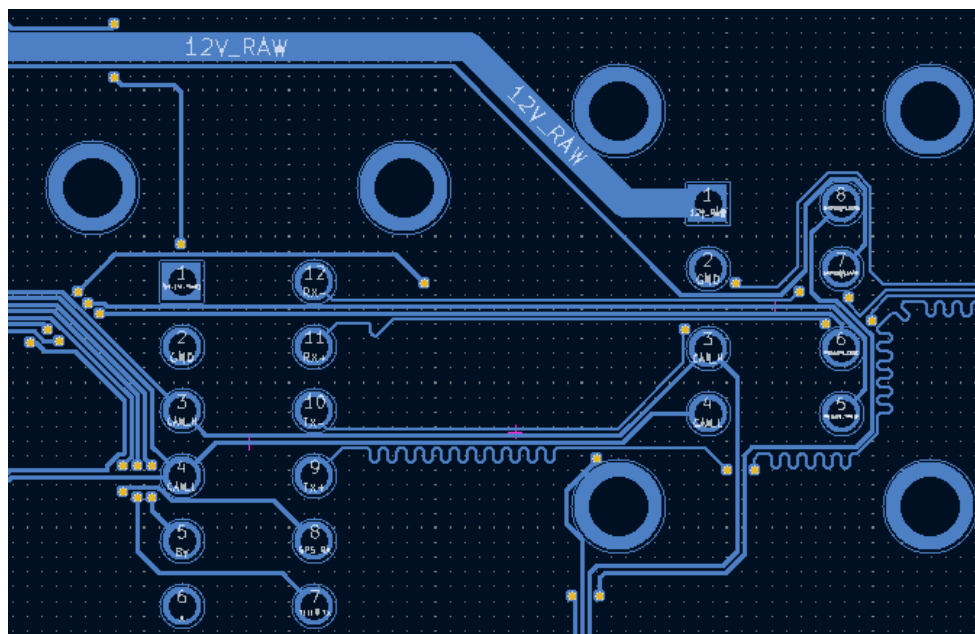


Figure 2: Effective trace routing on DEN V1 (NU23)

- Recognise the utility of **copper fill zones**. A filled zone does not have to encompass the entire board or be the GND net. It can be beneficial to include 12V and 5V zones or other commonly used nets to reduce routing. Zones do not have to follow edge cuts and use Ctrl + B to refreshes zones negating the need to redefine after changes.
- To improve **serviceability**, provide helpful information on the silkscreen to reduce ambiguity and create intuitive component recognition by displaying pinouts, functionality and proper test point names (*this will also aid commissioning and manufacture*). Figure 3 demonstrates effective use of silkscreen reducing the need to refer to KiCad and sift through other documentation to service the board. To avoid crowding and obscuring the silkscreen, check the 3D viewer often and examine the clarity from the perspective of another team member.

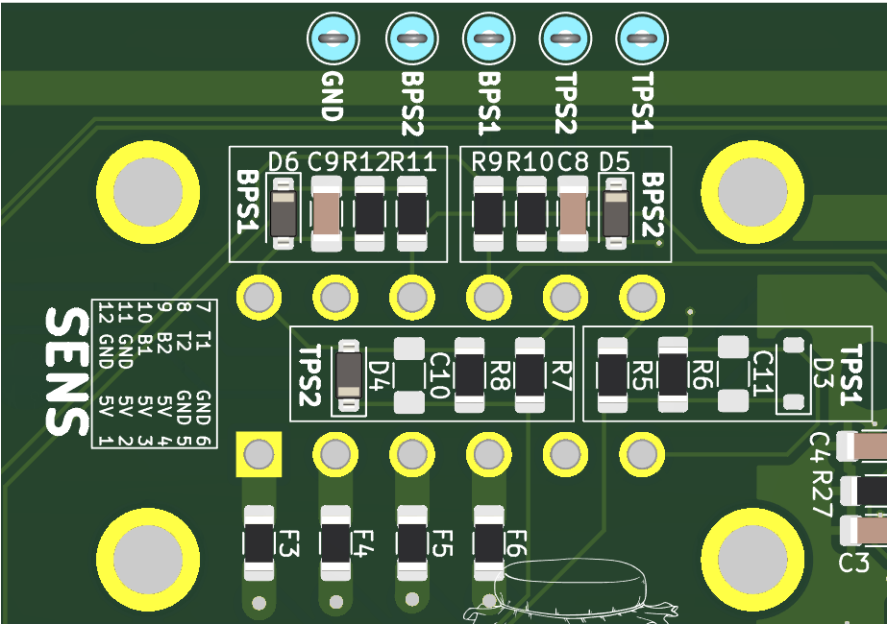


Figure 3: PEN V2 effective silkscreen labelling.

Prepare for Order

Detailed Design Review:

The DDR workbook states:

“The DDR Process is aimed towards reducing errors in PCB design and manufacture, and ensuring that the design intent of the PCB is in line with the requirements of the car and is compatible with existing systems.

This should be completed before any PCB is approved for manufacture.”

To respect the time of your peers, thoroughly **verify your own work** against the review criteria and fill out all required fields of the workbook before external review. Each individual section must be reviewed by the MCHA Department Lead before starting the next stage of design. Prior to submitting the PCB it must be checked by the MCHA Department Lead, Chief Engineer, and Malcolm Sidney.

- **PCB Design intent** – Keep this section accurate and concise so other engineers can quickly identify the functions and changes of your PCB.
- **Schematic Review** – All blocks should be either standard or validated in the research phase. Schematic should be easily readable with all crucial information present.
- **Layout Review** – Check the physical properties of the PCB, consider trace properties, limitations of PCB manufacturing and clear effective use of silkscreen elements.
- **CAD & Physical Design Review** – Ensure physical compatibility with the car and adhere to requirements from the mechanical design team (*team communication is key here*). Once the design is finalised, export the **STEP file** of the PCB to the “library and 3d parts” folder within the PCB directory (*PCB Design Zero-to-Hero -> Advanced Board Layout -> CAD Model Import/Export*). This is crucial for the mechanical team when designing PCB enclosures. When necessary, export a 3D printable version of the PCB to check the enclosure design. To do this, select (while holding ctrl) critical components in the layout editor (i.e. connectors, big relays, Teensy, screw terminals etc). Then export the STEP using only selected components (*File -> Export -> STEP -> Export Components = Only selected -> Export*)
- Remaining sheets can be completed after ordering or completing the PCB

Effective use of the DDR sheet is demonstrated by Figure 4, where notes can be used as reminders or improvement suggestions. For example, Figure 4 employs comments to track the assigned colour of different LEDs. Minimise the risk of design error by completing many DDRs with multiple experienced engineers.

Fusing / Protection			
Does the board have the necessary fusing/protection?	YES	YES	
Fuses on traces that power external devices?	NO	YES	Used PPTC fuses for sensor 5v supplies
Design Features			
GREEN LED for any OKHS passing through the board	NOT APPLICABLE	NOT APPLICABLE	
BLUE LED for any power and 'other' signals	YES	YES	12V_RAW, 12V_PROT, 5V, PRESSURE_OSCS, PRESSURE_THRESH
Log all Shutdown circuit states on microcontroller and RED LEDs	YES	YES	Using LED on board to show BOTS state
Screw Terminal (12V_RAW, GND) for commissioning (WJ126 Style 5mm Pitch)	YES	YES	

Figure 4: Effective use of DDR sheet and notes

Bill Of Materials:

To avoid delays, ensure the PCB components are stocked in the MCHA lab or ordered before the PCB arrives. This is done by completing a bill of materials (BOM). This process varies if you are using automated PCB assembly, refer to the *PCB Design Zero-to-Hero-> PCBA* video in this case.

- First **export the BOM** from KiCad by either using the JLCPCB plugin or *File -> Fabrication Outputs -> Bill of Materials*. Be aware if you intend to edit the BOM, save the changed version outside of the “productions” folder as this gets reset by KiCad when exporting.
- Ensure there is **sufficient stock** to populate 5 or 6 PCBs for most components. For larger, costly parts, use judgement to determine the level of redundancy required. For unique and costly parts, you may consider recycling components from outdated PCBs. If doing so, bench test the part, validating functionality and check with the department lead to ensure the PCB is no longer in use.
- For **low stock parts** identified, consider leaving excess stock available for other team members, redundancy and replacements. Place a PCB parts orders in Asana for depleted items.
- During BOM completion, check footprints once again by printing a 1:1 scale copy of the PCB. As done previously, verify correct footprint assignment but also identify any issues with the layout design and part proximity. By checking the scale version, it can be easier to find issues compared to a digital version.

Refer to Figure 5 for an example BOM. A BOM template is available at... *(work in progress, not yet available)*

Part	Value	QTY Per PCB	Quantity Needed	Stock	Footprint Fits?
TestPoint_Loop_D1.80mm_Drill1.0mm_Beaded		16	80	Sufficient	YES
1206 Resistor	19.6k	10	50	0	YES
DT15-12PA (12 pin DT)		1	2	4	YES
Potentiometer_Bourns_3296W_Vertical	10k	2	10	6	YES
5V Regulator: LM7805_DPAK		1	2	~20	YES

Figure 5: Example of completed BOM fields

PCB Order:

- Only order the PCB **once fully confident** in the design.
- Refer to *PCB Design Zero-to-Hero -> PCB Core Skills -> Export and Fabrication* for the ordering process. Ensure all PCB specifications are correct and confer with the department lead about any confusion. *Do not compromise hours of work with a simple error in the order process!*
- Log a purchase order in Asana for the PCB. Notify appropriate team members that the PCB is in the cart or that the Gerbers are ready for upload (depending on access permissions to the NU Teams PCB ordering accounts).

Commissioning:

Unit Test Checklist:

Before launching into soldering, prepare a unit test checklist to methodically plan your **commissioning procedure**. A template is available in the DDR workbook however employ any format that is most appropriate for you and your PCB (*i.e. a report or experiment method type document*). However, it is important to have this available to the team and to retain similar information fields and methodology. See the appendix for an example unit test checklist post commissioning.

This unit test checklist must include:

- A logical sequence of **manufacturing steps** and **subsequent tests** to guide someone through commissioning the entire PCB.
- After every **functional block** or testable circuit is manufactured, include a test step where it is verified as working before moving on. This **prevents backtracking** and reduces the difficulty of **fault isolation** on a fully populated board.
- Singular unit tests consisting of a **validation method** and **expected result**. These expected results must align with the original system design intent. At this stage of the design cycle, engineers should be fully comfortable describing how each part of their PCB operates. As a result, writing the checklist should only be knowledge consolidation.
- Test every combination of inputs and ensure all outputs have expected logic and voltages to protect downstream systems. *Test edge cases!*
- When commissioning multiples of the same board, include multiple result fields to correlate.

Unit Testing Importance:

The **time investment** of creating this checklist greatly reduces time spent commissioning and the risk of oversights. *You want to avoid discovering issues when the PCB is on the car or integrated with other systems and try to catch problems in isolation!* In reducing the cognitive load of using an improvised commissioning approach, more energy can be directed towards **quality work** and **sufficient validation**. Integrating an insufficiently tested PCB onto the competition platform not only risks damaging the PCB but also connected systems which expect specific voltages, currents and signals.

Manufacturing:

Ensure you are confident but not complacent when using the MCHA lab equipment. Adhere to all NU-Teams and Tunra Annexe rules and guidelines. Refer to the PCB Design and Lab Equipment Zero-to-Heros for safe and effective tooling techniques and always ask for help when uncertain. *Keep the MCHA Lab clean and organised!*

While referring to the pre-prepared unit test checklist, begin completing the **assemble, test, validate** cycle and update the checklist. Refer to the appendix for a completed example post manufacture. This cycle will materialise differently when using a stencil technique or when using automated PCB assembly where most parts are soldered at once. However, testing ideology remains the same.

Integration:

Once the PCB has been successfully commissioned, it is ready to be integrated into the wider electrical system of the competition platform. This is critical to validate not only the performance of your PCB in isolation but also its interaction with other systems. System level reliability and robustness take priority. These integration tests are planned in your in conjunction with fellow



engineers impacted by or dependent on your PCB. Remember to keep the mechanical team up to date as enclosure fitment will commence.

Methodically check functionality and use cases within the system as a bench test can not fully model real conditions. Employ the same level of scrutiny as the mid assembly validation and replicate the **expected results**.

Test **CANBUS interfaces** by using the kvaser leaf light v2 or CANdapter to read that intended signals are being sent and received.

Higher level integration tests will involve the entire car, and electrical systems. The integration procedure will end with on track testing where your design is slowly pushed to the limit. *You should be confident in your design given the diligent process you have followed!*

These track testing days provide opportune moments to validate with **real data** and evidence. Maximise this potential by monitoring all data channels relevant to your system. Watch for faults or **unintended behaviour**. Discuss with the data engineer or MoTeC manager about the possibility of adding more data streams and collecting additional information for your board. NU Teams strives for continuous improvement and better validation post design. Use this stage to confirm that your work meets real-world demands and to gather any final learnings or suggestions for future revisions.



PCB Maintenance and Service Life

After successfully deploying the PCB, your duties as the designer continue. Maintaining your PCB ensures lifelong performance and reliability. The ongoing responsibilities are as follows:

- Possible design iteration – depending on the results of unit testing and integration, future revisions of the PCB may be required. Implement the lessons learned and make use of all the rigorous documentation and note keeping you have conducted to track these concerns.
- Issue tracking – Be proactive in monitoring PCB or system level behaviour during testing and track days and promptly identify and document any irregularities.
- Documentation admin and upkeep – The attention to detail with PCB admin applied during previous stages will pay off when finalising documentation. Remember the importance of knowledge transfer when in the research phase of development. Begin preparing the necessary documentation for tech inspection and be ready to defend and justify your design.

The entire process of creating a competition qualified PCB requires discipline, diligence and attention to detail. In following this Zero-to-Hero, you have reduced the risk of compromising the competition platform and have successfully created a PCB for NU Racing. Be proud of your work and contributions to NU Racings success.

